The operator-algebra arc of the GeoVac framework: a reading guide for spectral triples, Latrémolière propinquity, and Lorentzian convergence

J. Loutey*

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Abstract

We provide a unified reading of the fourteen papers that make up the operator-algebra arc of the GeoVac framework (Papers 29, 32, 38, 39, 40, 42, 43, 44, 45, 46, 47, 48, 49, 50). The arc constructs an explicit almost-commutative spectral triple $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ on the Fock-projected unit three-sphere with a Camporesi–Higuchi spinor Dirac operator, places it in the Marcolli–van Suijlekom gauge-network lineage, and establishes four load-bearing results. First, the underlying truncated graphs are Ramanujan *at small sizes* — a finite-size graph-RH Observation (three of four families cross the Kotani–Sunada bound at modest sizes, Paper 29 §7), with integer-algebraic Ihara zeros (Paper 29). Second, the Connes–van Suijlekom truncations $\mathcal{T}_{n_{\max}}$ converge to the continuum spectral triple $\mathcal{T}_{S^3}$ in the van Suijlekom state-space Gromov–Hausdorff sense at quantitative rate $\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \leq C_3\gamma_{n_{\max}}$ with $C_3 = 1$ and asymptotic constant $4/\pi$ (Paper 38; dual-Coxeter normalisation of the SU(2) metric — $2/\pi$ on the unit S^3, the constant scaling with the metric, 2026-09-03). Third, the rate constant $4/\pi$ is universal across the class of compact connected semisimple Lie groups with bi-invariant metric — rigorous at rank 1, numerically pinned at ranks 1–2 for general G with the rank-uniform analytical proof a named gap — via a Plancherel-weight × Vandermonde-Jacobian cancellation in the dual-Coxeter normalisation (Paper 40). Fourth, the natural route to a Lorentzian convergence theorem on truncated $\mathrm{SU}(2) \times \mathrm{U}(1)_t$ Krein spectral triples — compressing to the Krein-positive subspace — is closed by a degeneracy theorem (Paper 45): the compression annihilates the spatial Dirac and the restricted Lipschitz seminorm vanishes identically. An earlier convergence claim through this device, together with the strong-form and bridge extensions built on it (Papers 46, 48, 49), is withdrawn; the Lorentzian analog of the convergence question Connes and van Suijlekom [7] deferred to "elsewhere" (and which Paper 38 closed on the Riemannian side) remains open, now with a sharpened specification. Paper 47's de-compactification limit $T \to \infty$ survives at the norm-resolvent / spectral level. A master-theorem *sketch* (§9) subsumes Papers 38, 39, and 40 as special cases: for arbitrary $k \geq 1$ and compact connected semisimple Lie groups $G_1, \ldots, G_k$, the k-fold tensor product converges (state-space GH) at the universal $4/\pi$ rate with Lipschitz comparison constant $C_3^{(k)} \leq \sqrt{k}$ (only the $k \leq 2$ cases are proved). The constant $4/\pi$ identifies, in both Riemannian (Paper 38) and Lorentzian (Paper 43 §10.2) settings, as the M_1 Hopf-base measure signature of the master Mellin engine catalogued by the case-exhaustion theorem of Paper 32 § VIII. Five further papers (48, 49, 50, 52, 53) extend the arc; Papers 48–50 are recorded in §10–§10.3, and Papers 52–53 are summarized at the end of this abstract; metric-level claims among them descoped by Paper 45's degeneracy theorem are reported here in corrected form. Paper 50 reproduces the Klebanov–Pufu–Safdi sphere F-theorem bit-exactly from the framework spectrum; Paper 52 frames the spectral-truncation convergence as a "Category III" discretization correspondence (the framework's analog of holography, at the state-space GH level); and Paper 53 treats the first non-homogeneous carrier — the disk-with-cone — where the finite disk is obstructed by its Dirichlet boundary

*Independent researcher. jloutey@gmail.com. This synthesis was prepared in an AI-augmented agentic workflow with Anthropic's Claude. All design decisions and editorial choices are the author's responsibility.

and convergence is recovered only on the boundaryless plane (Bochner–Riesz, rate $\Lambda^{-0.6}$ to $\Lambda^{-0.9}$). Paper 48 constructs Krein-pointed proper quantum metric spaces and a bridge functor to the Mondino–Sämann synthetic-Lorentzian pre-length spaces; the metric-level inheritance is descoped and the bridge survives as a conditional design. Paper 49 builds an OSLPLS (operator-system Lorentzian pre-length space) category and reads off-orbit superadditivity as the operator-algebraic dual of the twin paradox, with the deficit quantified by Datta's max-divergence chain inequality; the Λ-inheritance is descoped while the cocycle-deficit algebra survives. Paper 50 demonstrates bit-exact reproduction of CFT-on-sphere partition functions on S^3 and S^5 via the framework's existing Hurwitz machinery, revealing a master Mellin engine M_2/M_3 orthogonal decomposition of the (scalar, Dirac) sector and a log 3 cancellation theorem on S^5. These closures position the arc against the larger CFT/AdS lineage: the framework's substrate-level scale-freedom—packing diameter d_0 cancels out, scales enter only through the Fock focal length p_0—is the structural reflection of the scale-invariance that conformal field theory embodies at the physics level, and explains why the framework naturally hosts CFT-on-sphere observables bit-exactly. This synthesis gathers the statements, sketches the load-bearing proofs, and exhibits the structural relationships between the papers; no new theorems are introduced.

Status. The metric-level Lorentzian convergence claims of this arc are withdrawn: the K^+-compression device underlying Papers 45–46 and the bridge constructions of Papers 48–49 is degenerate (the compressed Lipschitz seminorm vanishes identically; Paper 45's annihilation theorem, §7). What stands: the unconditional Riemannian convergence theorem in van Suijlekom's state-space framework (Paper 38, translation-seminorm metrization), the operator-system substrate (Paper 44), the norm-resolvent de-compactification arrow (Paper 47), and the cocycle-deficit algebra and OSLPLS category design (Paper 49). Each affected section below carries a one-line status flag; the Status notes in the papers of record and the claims register (`docs/claims_register.md`) are authoritative for the full history.

1 Introduction

The GeoVac framework is a discretization scheme for separable quantum systems on natural geometries. When the geometry is the round $S^3 = \mathrm{SU}(2)$, the framework produces a sequence of finite truncations whose continuum limit is the Camporesi–Higuchi spectral triple of the round three-sphere [4]. The present paper synthesizes the nine documents in the project's operator-algebra group: Papers [30–38]. Together they form a self-contained research thread in mathematical operator algebras with four headline results.

Headline 1: the Hopf graphs are Ramanujan at small sizes. The finite Fock-projected S^3 Coulomb graphs and the Bargmann–Segal S^5 holomorphic graphs satisfy the Kotani–Sunada Ramanujan bound at small cutoffs — a *finite-size* graph-RH: three of four families (including S^3 Coulomb at $n_{\max} = 5$) cross the bound at modest sizes, so the property does not persist asymptotically (Paper 29 §7). Their *reciprocal* Ihara zeros (Hashimoto T-eigenvalues) are algebraic integers; the Ihara zeros themselves are algebraic over $\mathbb{Q}$ with integer-coefficient (but non-monic) minimal polynomials, and no π or transcendental content appears in the Ihara polynomial factorizations (Paper 29 [30], Observation [Ramanujan] and the closed-form Ihara factorizations of §5).

Headline 2: Riemannian state-space Gromov–Hausdorff convergence with explicit rate $4/\pi$. The Connes–van Suijlekom truncations $\mathcal{T}_{n_{\max}} = (\mathcal{O}_{n_{\max}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ of the SU(2) Camporesi–Higuchi spectral triple converge to the continuum triple $\mathcal{T}_{S^3}$ in the van Suijlekom state-space Gromov–Hausdorff sense [14, 15], with

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \;\leq\; C_3 \cdot \gamma_{n_{\max}}, \qquad C_3 = 1, \qquad \gamma_{n_{\max}} \sim \frac{4}{\pi}\frac{\log n_{\max}}{n_{\max}} \quad (n_{\max} \to \infty). \tag{1}$$

The asymptotic constant is the sphere-volume quotient $2\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3)$ at unit radius (numerically $\mathrm{Vol}(S^2)/\pi^2$) — the $k = 0$ volume factor of the master Mellin engine, not the Hopf fibration's base-to-total ratio (Paper 38, volume-ratio note; the fibration is $S^1 \to S^3 \to S^2$, corrected 2026-09-03) — and is discussed in detail in §4–§8 below (Paper 38 [32], Theorems 3.4 and 4.1).

Headline 3: the rate constant $4/\pi$ is universal across compact simple Lie groups. For any simple compact Lie group G with bi-invariant metric (so $G \in \{\mathrm{SU}(N), \mathrm{Spin}(n), \mathrm{Sp}(n), G_2, F_4, E_6, E_7, E_8\}$), the Plancherel-weight × Vandermonde-Jacobian cancellation theorem (Paper 40 §3.2 [34]) leaves exactly the rank-1 (SU(2)) constant $4/\pi$ of Paper 38 Appendix A from each positive root direction's Euler–Catalan asymptote, in the dual-Coxeter normalisation. The constant $C_3 = 1$ remains asymptotically tight at all ranks via the PRV-summand bound of Kumar (1988) and Vinberg (1990), applied to the dominant image of the tensor product $V_\lambda \otimes V^*_{\lambda'}$. *Tier:* this universality is rigorous at rank 1 (SU(2)) and, for general G, is a proof-sketch mechanism (the Plancherel×Vandermonde cancellation) numerically pinned at ranks 1–2 $(\mathrm{SU}(2), \mathrm{SU}(3), \mathrm{Sp}(2), G_2)$ with one rank-3 datapoint; a fully rank-uniform analytical proof is a named gap (Paper 40 §L2), and the per-group rate-constant extractions are fit-sensitive.

Headline 4: the Lorentzian propinquity route is closed by a degeneracy theorem. On truncated $\mathrm{SU}(2)\times\mathrm{U}(1)_T$ Krein spectral triples $\mathcal{T}^L_{n_{\max},N_t,T}$ built via the Bizi–Brouder–Besnard [2] signature $(3,1)$ data at $(m,n) = (4,6)$, the natural K^+-restricted weak-form Lorentzian propinquity target bound is

$$\Lambda^L(\mathcal{T}^L_{n_{\max},N_t,T}, \mathcal{T}^L_{\mathcal{M}}) \;\leq\; C_3^{\mathrm{joint}} \cdot \gamma^{\mathrm{joint}}_{n_{\max},N_t,T} \;\longrightarrow\; 0, \tag{2}$$

with $\gamma^{\mathrm{joint}}_{n_{\max},N_t,T} = O(\log n_{\max}/n_{\max} + T/N_t)$, where $\Lambda^L(\cdot,\cdot) := \Lambda(\cdot|_{\mathcal{K}^+}, \cdot|_{\mathcal{K}^+})$ is the propinquity of the standard metric spectral triple obtained from the Krein-positive subspace $\mathcal{K}^+ = \{|\psi\rangle : J_L|\psi\rangle = +|\psi\rangle\}$. The target fails structurally: the quantity Λ^L is degenerate — the K^+ compression annihilates the spatial Dirac and the restricted Lipschitz seminorm vanishes identically (Paper 45's annihilation theorem [38]) — and the earlier convergence claim through this device (formerly Theorem 5.1 of Paper 45) is withdrawn. The Lorentzian analog of the convergence question Connes and van Suijlekom [7] deferred to "elsewhere" (closed on the Riemannian side by Paper 38) remains open, now with a sharpened specification.

Position against the literature. The arc sits in the lineage of Connes–van Suijlekom 2021 [7], which formalised spectral truncations as operator systems with finite-cutoff Connes distance; of Marcolli and van Suijlekom 2014 [17], which introduced gauge networks as a discrete-graph realisation of finite spectral triples carrying connection on edges; and of the subsequent Latrémolière–Hekkelman–McDonald–Toyota–Farsi–Leimbach school of quantitative propinquity convergence for spectral truncations [8, 9, 12–14, 16, 26]. All these earlier convergence results are strictly Riemannian (and almost all are on flat structures S^1, $\mathbb{T}^d$, S^2 Berezin–Toeplitz). The Paper 38 SU(2) result and the Paper 40 generalisation are non-abelian non-flat instances at quantitative rate (unconditional as of June 2026, via the translation-seminorm metrization; scalar-sector convergence for compact groups is due to Gaudillot-Estrada and van Suijlekom, arXiv:2310.14733); on the Lorentzian side the Paper 45 contribution is a degeneracy theorem, not a convergence instance. Mondino and Sämann's synthetic Lorentzian Gromov–Hausdorff program [18, 19] is categorically distinct in object class (pre-length spaces with causal diamonds, not operator-algebraic spectral triples) and is not in competition with the present arc.

What this synthesis provides. A working mathematician reading the corpus cold may find the nine-paper sequence difficult to navigate. This document collects the constructions, restates the load-bearing theorems with cross-references back to the original papers, and exhibits the relationships between them. No new theorems are introduced. The reader is assumed familiar with spectral triples [5, 6], Lipschitz seminorms and Rieffel quantum metric spaces, and the Latrémolière propinquity [14, 15]. Familiarity with [7] is helpful but not strictly required. No knowledge of the GeoVac physics-side corpus (quantum chemistry, QED, precision spectroscopy) is assumed; this arc is self-contained from the operator-algebra side.

2 The GeoVac spectral triple

2.1 The Camporesi–Higuchi data on S^3

We work throughout on the unit three-sphere S^3, identified with SU(2) under its bi-invariant round metric. Following Camporesi and Higuchi [4], the spinor bundle is parallelisable, of rank 2, and the Dirac operator D_{CH} has spectrum

$$\mathrm{spec}(D_{\mathrm{CH}}) = \{\pm(n+\tfrac{3}{2}) : n \in \mathbb{N}_0\}, \qquad g_n^{\mathrm{Dirac}} = 2(n+1)(n+2) \tag{3}$$

on each chirality of the half-integer Hurwitz-shifted shell. We write $\mathcal{H}_{S^3} = L^2(S^3, \Sigma)$ for the chirality- doubled spinor Hilbert space carrying the Camporesi–Higuchi data.

2.2 The GeoVac algebra, Hilbert space, and Dirac operator

The GeoVac construction (Paper 32 §3 [31]) takes the following data:

- $\mathcal{A}_{\mathrm{GV}} \subset C^\infty(S^3)$ is the involutive algebra of complex-valued smooth functions on S^3, with the standard pointwise product and complex conjugation;
- $\mathcal{H}_{\mathrm{GV}} = \mathcal{H}_{S^3}$, with $\mathcal{A}_{\mathrm{GV}}$ acting by pointwise multiplication;
- $D_{\mathrm{GV}} = D_{\mathrm{CH}}$, the Camporesi–Higuchi Dirac operator;
- J_{GV} is the charge-conjugation real structure inherited from the spin structure of S^3, satisfying $J_{\mathrm{GV}}^2 = -I$ (Euclidean KO-dimension 3), $J_{\mathrm{GV}} D_{\mathrm{GV}} = +D_{\mathrm{GV}} J_{\mathrm{GV}}$.

The triple $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}}; J_{\mathrm{GV}})$ is real, odd (no $\mathbb{Z}_2$-grading; KO-dimension 3), and has compact resolvent $|D_{\mathrm{GV}}|^{-1}$ with eigenvalue multiplicities given by (3). The Connes axiom audit is recorded in Paper 32 §4 [31], where each of the seven axioms (order zero, bounded commutators, compact resolvent, order one, real structure, $\mathbb{Z}_2$-grading by convention, and finite summability) is checked.

2.3 The Marcolli–van Suijlekom lineage

The role of GeoVac in the ecosystem of finite-and-graph spectral triples is explained in Paper 32 §7. The Marcolli–van Suijlekom construction [17] attaches a finite spectral triple to each vertex of a graph and a connection (gauge group element) to each edge; the spectral action of the resulting object recovers Wilson lattice gauge theory. The continuum-limit issue addressed by Perez-Sanchez [21] clarifies that the limit is Yang–Mills without Higgs by default; a Higgs sector appears only via off-diagonal data on the finite factor.

The GeoVac Fock-projected S^3 graph fits this template naturally. Its vertices are the Fock index (n, ℓ, m) with $n = 2j + 1$, ℓ the orbital, and m the magnetic quantum number, providing a Peter–Weyl-indexed basis on $S^3 = \mathrm{SU}(2)$. Each Fock shell carries a finite spectral triple (roughly, a finite-dimensional subrepresentation of the Camporesi–Higuchi triple), and the edge structure encodes the SO(4) selection rules of the underlying graph. The graph admits a $U(1)$ Wilson construction (recorded as GeoVac internal Paper 25), an SU(2) Wilson construction on

$S^3 = \mathrm{SU}(2)$ (Paper 30), and an SU(3) Wilson construction on the Bargmann–Segal S^5 [29], all of which carry the universal kinetic coefficient $1/(4N_c)$.

The Connes axiom audit (Paper 32 §4 [31]) checks explicitly that $J^2 = -I$ and $JD = +DJ$ hold on the truthful Camporesi–Higuchi data at every finite cutoff $n_{\max} \in \{1, 2, 3\}$ tested. These two relations, together with the parallelisability of $S^3 = \mathrm{SU}(2)$, distinguish the GeoVac triple from the Connes–van Suijlekom circulant comparator, which is commutative and gives propagation number 1.

2.4 Truncations as operator systems

For each $n_{\max} \in \mathbb{N}$, the orthogonal projection $P_{n_{\max}} : \mathcal{H}_{\mathrm{GV}} \to \mathcal{H}_{\mathrm{GV}}$ onto the span of Peter–Weyl modes with $n \leq n_{\max}$ defines a truncated operator system

$$\mathcal{O}_{n_{\max}} \; := \; P_{n_{\max}} \mathcal{A}_{\mathrm{GV}} P_{n_{\max}} \; \subset \; \mathcal{B}(P_{n_{\max}} \mathcal{H}_{\mathrm{GV}}), \tag{4}$$

which is self-adjoint, contains the identity, but is in general *not* multiplicatively closed. The defect is precisely the content of the Connes–van Suijlekom operator-system formalism [7]: in the natural envelope, $\mathcal{O}_{n_{\max}}$ has propagation number $\mathrm{prop}(\mathcal{O}_{n_{\max}}) = 2$ at every tested $n_{\max} \in \{2, 3, 4\}$, matching the Toeplitz S^1 result [7, Proposition 4.2] verbatim. A concrete witness pair at $n_{\max} = 2$ is $M^2_{2,1,0} \notin \mathcal{O}_2$ with 14.9% Frobenius residual against $\mathcal{O}_2$, deficit concentrated on the top-shell diagonal in the natural Avery basis (Paper 32 §3.4 [31]). The truncated triple is then

$$\mathcal{T}_{n_{\max}} = (\mathcal{O}_{n_{\max}}, P_{n_{\max}} \mathcal{H}_{\mathrm{GV}}, P_{n_{\max}} D_{\mathrm{GV}} P_{n_{\max}}),$$

with continuum limit $\mathcal{T}_{S^3} = (\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$.

3 The Hopf graphs as Ramanujan substrate

Before turning to convergence, we record the most basic spectral-graph property of the underlying finite truncations.

Ihara zeta of the Fock graph. For each $n_{\max}$, let $G_{n_{\max}}$ denote the directed Fock-projected S^3 Coulomb graph: vertices are triples (n, ℓ, m) with $n \leq n_{\max}$, $\ell < n$, $|m| \leq \ell$; edges encode SO(4) selection rules. The Ihara zeta function $\zeta_{G_{n_{\max}}}(s)$ is computed in Paper 29 §4 [30] via the Ihara–Bass determinantal formula and via the $2E \times 2E$ Hashimoto non-backtracking edge operator.

Remark 3.1 (Observation — Paper 29, Ramanujan bound + §7 bound-crossing)**.** At small cutoffs the Fock-projected S^3 graph $G_{n_{\max}}$ satisfies the Kotani–Sunada Ramanujan bound strictly: the largest non-trivial Hashimoto eigenvalue sits strictly inside the spectral radius $\sqrt{q_{\max}}$ associated with the maximum (Perron) vertex degree. This is a *finite-size* statement, not an asymptotic theorem: Paper 29 §7 shows three of four families (including S^3 Coulomb at $n_{\max} = 5$) *cross* the bound at modest sizes, so the graph-RH does not persist as $n_{\max} \to \infty$. The reciprocal Ihara zeros (Hashimoto T-eigenvalues) are algebraic integers, while the Ihara zeros themselves are algebraic over $\mathbb{Q}$; the corresponding Ihara polynomials factor over $\mathbb{Z}[s]$ into integer-coefficient pieces, and several factorizations admit explicit closed forms (e.g. $(4s^2 + 1)^2$, $(9s^2 + 1)^4$, the $12 + 22$ dichotomy for S^5 at $N_{\max} = 3$).

Structural consequences. The Observation 3.1 secures two facts about the truncated graphs that are used implicitly elsewhere in the arc. First, no π or transcendental quantity appears anywhere in the Ihara polynomials, in any of the tested factorizations (Paper 29 §5.4 [30], Corollary on integer-algebraicity). This is the graph-spectral analog of the π-source case-exhaustion theorem (Paper 32 §8): both say that π enters only through specific named mechanisms (Hopf-base measure, spectral action, vertex-parity Hurwitz character), never in the bare combinatorial data. Second, the Hashimoto-spectrum Poisson statistics (Paper 29 §6.3) are categorically distinct from the GUE-like statistics seen on the spectral-zeta side of the framework, making clear that the Weil dictionary between Selberg and Riemann breaks at GeoVac by design.

The Ramanujan property of the truncated graphs underwrites our freedom to take naive sums and traces over $G_{n_{\max}}$ without worrying about pathological spectral gaps: the cutoff graphs are well-conditioned for spectral analysis, and the Ihara polynomials are entirely in $\mathbb{Z}[s]$. This is exactly the kind of substrate one wants under any convergence theorem to follow.

4 Riemannian Gromov–Hausdorff convergence: the central theorem

4.1 Statement (Paper 38)

The load-bearing theorem of the arc is Paper 38 [32], Theorem 3.4.

Theorem 4.1 (Paper 38, GH convergence on $S^3 = \mathrm{SU}(2)$). *The Connes–van Suijlekom truncations $\mathcal{T}_{n_{\max}}$ of the round $\mathrm{SU}(2)$ Camporesi–Higuchi spectral triple converge to $\mathcal{T}_{S^3}$ in the van Suijlekom state-space Gromov–Hausdorff sense:*

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \;\leq\; C_3 \cdot \gamma_{n_{\max}} \;\to\; 0 \quad (n_{\max} \to \infty), \tag{5}$$

with $C_3 = 1$ (asymptotic tight via the PRV summand bound) and asymptotic constant $4/\pi$:

$$n\,\gamma_n \;\sim\; (4/\pi)\,\log n \qquad (n \to \infty). \tag{6}$$

A uniform finite-cutoff bound $\gamma_n \leq 6\,(\log n)/n$ holds for all $n \geq 2$.

The proof proceeds via five lemmas L1′, L2, L3, L4, L5 of Paper 38 §4–§6. We sketch each.

4.2 Lemma L1′: chirality-doubled truncated operator system

The original Connes–van Suijlekom truncation is on the scalar Fock graph; the spinor lift to the full Camporesi–Higuchi bundle (with both chiralities) is the subject of an internal sprint (R3.5).[1] The truthful Camporesi–Higuchi data at finite $n_{\max}$ has the property that on most cross-shell pairs the Connes distance is $+\infty$ (an artefact of the n-shell degeneracy in the Avery basis); this is repaired by lifting to the full Dirac sector with both chiralities and off-diagonal shell-coupling multipliers. The chirality-doubled operator system has dimension $\dim \mathcal{O}_{n_{\max}} = 1, 14, 55, 140, \ldots$ at $n_{\max} = 1, 2, 3, 4, \ldots$ and every cross-shell pair has finite Connes distance. See Paper 38 §4.1 [32] for the explicit construction.

4.3 Lemma L2: central spectral Fejér kernel on $\mathrm{SU}(2)$

The propinquity tunneling pair built in L5 below requires a positive, state-preserving, central good kernel on SU(2) whose mass-concentration moment vanishes at a known rate. Paper 38 §4.2 constructs the central spectral Fejér kernel

$$K_{n_{\max}}(g) \;=\; \frac{1}{Z_{n_{\max}}} \left| \sum_{j \leq j_{\max}} \sqrt{2j+1}\,\chi_j(g) \right|^2, \qquad j_{\max} = (n_{\max}-1)/2, \quad Z_{n_{\max}} = \frac{n_{\max}(n_{\max}+1)}{2}. \tag{7}$$

[1] Parenthetical sprint/track designations are anchors into the project's internal chronicle (`CHANGELOG.md` in the repository); "WHn" designations are entries of the internal working-hypothesis register (`CLAUDE.md` §1.7).

The Plancherel symbol on the Avery–Wen–Avery basis is $\widehat{K}(j) = (2j+1)/Z_{n_{\max}}$ for $j \le j_{\max}$. The cb-norm of the associated Herz–Schur multiplier acting on the group C^*-algebra is

$$\|S_{K_{n_{\max}}}\|_{\mathrm{cb}} \;=\; \frac{2}{n_{\max}+1} \;=\; O(1/n_{\max}), \tag{8}$$

via the Bożejko–Fendler central-multiplier equality [3] on amenable compact groups (Pier [22]; Paulsen [20]; Pisier [23]).

The asymptotic constant $4/\pi$. The mass-concentration moment $\gamma_{n_{\max}}$ of the kernel admits an exact closed-form sum rule whose asymptotics are fixed by an Euler–Maclaurin analysis of two logarithmic sub-coefficients (net $2-1=1$); the corpus names this the *Stein–Weiss sharpening*, after the real-line approximate-identity estimates of [25] (Paper 38, Appendix A):

$$\gamma_{n_{\max}} \;=\; \pi - \frac{4\,T_{n_{\max}}}{\pi\,Z_{n_{\max}}}, \quad \text{with } \frac{n\,\gamma_n}{\log n} \to \frac{4}{\pi} \ \text{ as } n \to \infty. \tag{9}$$

The constant is the sphere-volume quotient $\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3) = 2/\pi$ at unit radius, twice that in the dual-Coxeter normalisation. Both spheres have radius 1, so this is *not* the Hopf fibration's base-to-total ratio (that base is $S^2(1/2)$ and the ratio is $1/(2\pi)$; Paper 38, volume-ratio note, 2026-09-03). Concretely $\mathrm{Vol}(S^2) = 4\pi$ and $\pi^2 = \mathrm{Vol}_1(\mathrm{SU}(2))/2$, so that $4/\pi = 2\mathrm{Vol}(S^2)/\mathrm{Vol}_1(\mathrm{SU}(2))$ in the dual-Coxeter normalisation; on the unit S^3 the constant is the plain Haar ratio $\mathrm{Vol}(S^2)/\mathrm{Vol}(\mathrm{SU}(2)) = 2/\pi$, the factor 2 being the metric scale (Paper 38, 2026-09-03). The classical Fejér kernel on $\mathbb{T}^1$ has probability-normalised first-moment constant $2/\pi$ — half the SU(2) value in the dual-Coxeter normalisation and equal to it on the unit S^3, so the factor 2 is the metric scale rather than a signature of the group (Paper 38, Remark "Connection to the circle Fejér estimate"; re-priced 2026-09-03). The expected one-log slowdown of the non-abelian SU(2) rate compared with the Leimbach–van Suijlekom torus rate $O(1/n_{\max})$ [16] is precisely the $\log n$ factor in (9).

4.4 Lemma L3: Lipschitz comparison with $C_3 = 1$

For each $f \in C^\infty(S^3)$ and each truncated multiplier M_f on $\mathcal{O}_{n_{\max}}$, the Lipschitz comparison

$$\|[D_{\mathrm{CH}}, M_f]\|_{\mathrm{op}} \;\le\; C_3 \cdot \|\nabla f\|_{L^\infty(S^3)} \tag{10}$$

holds with $C_3 = 1$ (Paper 38, Theorem 5.2). The proof uses SO(4) selection rules: the off-diagonal element of $[D_{\mathrm{CH}}, M_f]$ between Peter–Weyl shells V_λ and $V_{\lambda'}$ is bounded by $\sqrt{C(\lambda)} - \sqrt{C(\lambda')}$ where $C(\lambda)$ is the quadratic Casimir of the highest weight; the Dirac-triangle inequality $|D(\pi) - D(\pi')| \le \sqrt{C(\sigma)}$ on intertwining summands $\sigma \subset \pi \otimes \pi'^*$ secures $C_3 \le 1$, and the Avery harmonic gradient scaling forces $C_3 = 1$ in the limit $n_{\max} \to \infty$. At any finite $n_{\max}$, $C_3 = (N-1)/\sqrt{N^2-1}$ for $N = n_{\max}$, which converges to 1 from below.

The same bound is proved for all simple compact Lie groups in Paper 40 §3.3 [34], with asymptotic tightness $C_3(G) = 1$ at all ranks; see §5 below.

4.5 Lemma L4: Berezin reconstruction

A Berezin map $B_{n_{\max}} : C(S^3) \to \mathcal{O}_{n_{\max}}$ is constructed in Paper 38 §5.3 as

$$B_{n_{\max}}(f) \;=\; \sum_{N \le n_{\max},\, L,\, M} \widehat{K}(N)\, c_{NLM}\, M_{NLM}, \quad \widehat{K}(N) = N/Z_{n_{\max}}. \tag{11}$$

This is the SU(2) non-Kähler analog of the Connes–van Suijlekom [7] Fejér kernel reconstruction on S^1 and the non-holomorphic analog of Hawkins [11]. Four properties are established:

(a) Positivity: for $f \geq 0$, $B_{n_{\max}}(f)$ is a positive operator on $P_{n_{\max}}\mathcal{H}_{\mathrm{GV}}$ (via convolution + positivity-preserving compression);

(b) Contractivity: $\|B_{n_{\max}}(f)\|_{\mathrm{op}} \leq \|f\|_\infty$;

(c) Approximate identity: $\|B_{n_{\max}}(f) - P_{n_{\max}} M_f P_{n_{\max}}\|_{\mathrm{op}} \leq \gamma_{n_{\max}} \|\nabla f\|_\infty$ with $\gamma_{n_{\max}}$ as in L2;

(d) L3 compatibility: $\|[D_{\mathrm{CH}}, B_{n_{\max}}(f)]\|_{\mathrm{op}} \leq \|\nabla f\|_\infty$ (i.e. $C_3 = 1$ is inherited via $\widehat{K} \leq 1$).

This is the operator-system analog of the Berezin–Toeplitz quantisation; see Connes–van Suijlekom [7] for the operator-system truncation of the closely related $\mathbb{T}^d$ case.

4.6 Lemma L5: Latrémolière propinquity assembly

The tunneling pair $(B_{n_{\max}}, P_{n_{\max}})$ between $\mathcal{T}_{S^3}$ and $\mathcal{T}_{n_{\max}}$ has reach $\leq \gamma_{n_{\max}}$ (by L4(c) approximate identity and L2(g) Bożejko–Fendler dual estimate), and height contributions are bounded by L4(b) contractivity and the fact that the projection P contributes $\mathrm{height}_P = 0$. The assembly machinery is Latrémolière [14], applied verbatim. This gives

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \; \leq \; C_3 \cdot \gamma_{n_{\max}} \; \to \; 0, \tag{12}$$

completing the proof of Theorem 4.1.

4.7 Numerical verification

Paper 38 records the explicit Λ values at small cutoffs:

$$\Lambda(\mathcal{T}_2, \mathcal{T}_{S^3}) \approx 2.0746, \quad \Lambda(\mathcal{T}_3, \mathcal{T}_{S^3}) \approx 1.6101, \quad \Lambda(\mathcal{T}_4, \mathcal{T}_{S^3}) \approx 1.3223, \tag{13}$$

with $\Lambda(4)/\Lambda(2) = 0.637$. These are monotone decreasing and consistent with the $O(\log n_{\max}/n_{\max})$ rate.

4.8 Tensor extension (Paper 39)

Paper 39 extends Theorem 4.1 to tensor products $\mathcal{T}_{S^3}^{\lambda_a} \otimes \mathcal{T}_{S^3}^{\lambda_b}$ of two copies of the same manifold at distinct focal lengths $\lambda_a \neq \lambda_b$ [33]. The result is a proved-at-qualitative-rate theorem closing the published-open NCG question of convergence for the tensor of two infinite metric spectral triples. Two technical points underwrite the closure:

- **R1: tensor Lipschitz constant via the tight triangle bound.** The graded Leibniz rule (Connes–Marcolli form) splits the joint commutator into two terms; the operator-norm *triangle* bound on that sum gives the full operator-system constant $C_3^{(2)} \leq \sqrt{2}$ (increasing to $\sqrt{2}$). *Correction (2026-06-18):* an earlier draft claimed a *Pythagorean* refinement $C_3^{(2)} < 1$; that identity is false (graded anticommutation does not give operator-norm orthogonality — verified maximally violated on the actual harmonics), and the triangle bound is tight. Convergence is unaffected ($\sqrt{2}$ is a finite constant); only the cosmetic "< 1" sharpness claim changed.
- **R2: Latrémolière 2017 explicit $\varepsilon_{\mathrm{cross}}$.** The cross term in the tunneling pair is bounded explicitly by Latrémolière [14] §4, with rate $O(\max \gamma)$ and constant $\leq 2\sqrt{2}$.

The bit-identical convergence ratio $\Lambda(n_{\max}, n_{\max})/\Lambda(2, 2)$ on the diagonal matches the single-factor Paper 38 ratio numerically, providing a load-bearing sanity check.

5 The universality of the rate constant $4/\pi$

Paper 40 [34] extends Theorem 4.1 from SU(2) to the class of all simple compact Lie groups with bi-invariant metric.

Theorem 5.1 (Paper 40, universal rate constant)**.** *Let G be a compact connected semisimple Lie group of rank $r \geq 1$ with bi-invariant metric (each simple factor $\in \{\mathrm{SU}(N), \mathrm{Spin}(n), \mathrm{Sp}(n), G_2, F_4, E_6, E_7, E_8\}$). The truncated Camporesi–Higuchi triples $\mathcal{T}_{n_{\max}}(G)$ converge to $\mathcal{T}_G$ in the van Suijlekom state-space Gromov–Hausdorff sense with the same asymptotic rate constant as (6): namely $4/\pi$, in the dual-Coxeter normalisation. The constant $C_3(G) = 1$ remains asymptotically tight at all ranks. For general G this is conditional on a per-group spin-window decomposition (verified for* SU(2)*).*

Plancherel-weight $\times$ Vandermonde-Jacobian cancellation. The proof of Theorem 5.1 runs through a structural identity which we record because it is the deepest algebraic content of the universality. In the central spectral Fejér kernel on G,

$$K_{n_{\max}}(g) \;=\; \frac{1}{Z_{n_{\max}}(G)} \left| \sum_{\lambda\,:\,|\lambda+\rho|^2 \leq M(n_{\max})^2} \sqrt{\dim V_\lambda}\, \chi_\lambda(g) \right|^2, \tag{14}$$

the Plancherel weight $\sqrt{\dim V_\lambda}$ on the squared amplitude (so the weight is $\dim V_\lambda$) and the Vandermonde Jacobian $|\Delta(t)|^2$ in the Weyl integration formula on G share the same $\prod_{\alpha>0}$ product structure over positive roots. In the dual-Coxeter normalisation these products cancel exactly, leaving only the rank-1 (SU(2)) constant $4/\pi$ of Paper 38 Appendix A from each positive-root direction's Euler–Catalan asymptote $\sum_{d\text{odd}} 1/d^2 = \pi^2/8$. The universal constant $4/\pi$ is then *independent* of rank, Weyl group order, and Lie type.

Asymptotic tightness $C_3(G) = 1$ via PRV. The Lipschitz comparison bound (10) extends to G via the PRV-summand theorem (Kumar 1988, Vinberg 1990): for every Weyl group element $w \in W$, the dominant image σ_w of $\lambda + w\lambda'^*$ is a summand of $V_\lambda \otimes V_{\lambda'}^*$, and $|\sigma_w + \rho|^2 \geq |\lambda - \lambda'|^2 + |\rho|^2$. This covers the asymptotic family $(n_{\max}, 0, \ldots, 0) \otimes (1, 0, \ldots, 0)^*$ rigorously, giving $C_3(G) = 1$ asymptotic-tight in all simple Lie algebra types $A_n, B_n, C_n, D_n, G_2, F_4, E_6, E_7, E_8$. Interior (non-PRV) summands at finite cutoff are verified numerically across the panel $\mathrm{SU}(3), \mathrm{SU}(4), \mathrm{Sp}(2), G_2$ at 5641/5641 cells; analytic closure via Brauer–Klimyk signed-sum cancellation is bookkeeping.

Companion: state-space GH. The state-space Gromov–Hausdorff convergence (without rate and without Dirac) at the same generality was proved by Gaudillot-Estrada–van Suijlekom (IMRN 2025, paper rnaf197). Paper 40 provides the missing rate constant, the Lipschitz spinor bound, and the asymptotic tightness on the state-space GH substrate; the further step from state-space GH to spectral-triple-level Latrémolière propinquity (the dual-reach step) is a conceded named gap (the same scope boundary as Paper 38), so for general G the result is conditional on a per-group spin-window decomposition, verified here only for SU(2).

6 Modular structure and Wick rotation

6.1 The four-witness Wick-rotation theorem (Paper 42)

The Wick rotation between Euclidean and Lorentzian field theory is classically formulated through four equivalent presentations on a flat spacetime:

- Hartle–Hawking 1976 [10]: the temperature of the Euclidean cigar's τ-circumference identifies as the Hawking temperature;

- Sewell 1982 [24]: the KMS state at inverse-temperature $\beta = 2\pi/\kappa_g$ is the wedge vacuum;
- Bisognano–Wichmann 1976 [1]: the modular automorphism group of the wedge von Neumann algebra acts as boosts around the wedge edge;
- Unruh 1976 [27]: an accelerated observer with proper acceleration a sees a $T_U = a/(2\pi)$ thermal bath in the vacuum.

On a flat background, the four are equivalent under standard Wick-rotation arguments. On a curved background like S^3 they are not obviously equivalent, and at finite cutoff (operator-system level) they require separate construction.

Paper 42 [35] establishes the four-witness theorem at the operator-system level on the truncated Camporesi–Higuchi spectral triple $\mathcal{T}_{n_{\max}}$ via two structurally independent but operator-action conjugate constructions:

Theorem 6.1 (Paper 42, BW-α geometric realisation)**.** *For each $n_{\max} \in \{2, 3, 4, 5\}$, there exists a wedge projection $P_W = \frac{1}{2}(I + R_{\text{polar}})$ on $P_{n_{\max}}\mathcal{H}_{\text{GV}}$ (bit-exact idempotent and Hermitian) and a geometric modular Hamiltonian*

$$K_\alpha^W \;=\; J_{\text{polar}} \quad \textit{(rotation generator preserving } W\textit{)}, \tag{15}$$

with integer spectrum $\{\underline{\text{two}\, m_j}\}$ on the full-Dirac basis. The modular flow satisfies $\sigma_{2\pi}(O) = O$ bit-exact (residual $O(\sqrt{\dim \mathcal{H}} \cdot \varepsilon_{\text{machine}})$) for the four witnesses BW, Hartle–Hawking, Sewell, Unruh, with cross-witness collapse residual identically zero.

Theorem 6.2 (Paper 42, BW-γ Tomita–Takesaki realisation)**.** *On the Krein-GNS Hilbert–Schmidt space $\mathcal{H}_{\text{GNS}} = M_{\dim W}(\mathbb{C})$ with KMS state $\rho_W = e^{-K_\alpha^W}/Z$ and the BW choice $H_{\text{local}} := K_\alpha^W/\beta$, the polar decomposition $S = J_{\text{TT}} \cdot \Delta^{1/2}$ gives a Tomita-Takesaki modular Hamiltonian $K_{\text{TT}} = -\log \Delta$ with the same integer spectrum, and the operator-action conjugacy*

$$\sigma_t^{\text{TT}}(a) \;=\; \sigma_{-t}^{\alpha}(a) \tag{16}$$

holds bit-exact at every tested $t \in \{1, \pi, 2\pi\}$. The modular operator Δ satisfies $J_{\text{TT}}^2 = +I$, categorically distinct from the Connes J_{GV} (KO-dim 3, $J^2 = -I$).

Load-bearing scope finding (Paper 42 §7.2). A key derived structural finding is that the canonical local Hamiltonian for the $\beta = 2\pi$ Tomita–Takesaki KMS state is $H_{\text{local}} = K_\alpha^W/\beta$ rather than the intrinsic Camporesi–Higuchi Dirac operator D_{CH}. The intrinsic D_{CH} has half-integer spectrum, so its operator-level period lift is $e^{i \cdot 2\pi \cdot D_{\text{CH}}} = -I$ (the spinor double-cover sign), whereas the odd-integer K_α^W lifts to $+I$. The *conjugation* modular flow $\sigma_{2\pi}$ closes with either generator (a global scalar cancels in the conjugation), so the distinction is at the operator level, not the flow-closure level: only K_α^W lifts the period to the identity operator. Spectral-action (Chamseddine–Connes [5]) and thermal-time (Connes–Rovelli) paradigms point to different generators on the wedge KMS state. This is the open question O3 of Paper 42 §10, sharpened in Paper 43 to the signature-independence statement below.

6.2 Lorentzian extension at finite cutoff (Paper 43)

Paper 43 [36] extends the four-witness theorem to the Lorentzian Krein space at signature (3, 1), built via the Bizi–Brouder–Besnard prescription [2] at $(m, n) = (4, 6)$ in the Peskin–Schroeder chiral basis, with Lorentzian Dirac

$$D_L \;=\; i\left(\gamma^0 \otimes \partial_t \;+\; D_{\text{GV}} \otimes I_{N_t}\right) \tag{17}$$

per van den Dungen 2016 Proposition 4.1 [28].

Theorem 6.3 (Paper 43 §5–§7, Lorentzian four-witness at finite cutoff)**.** *At every tested* $(n_{\max}, N_t) \in \{1,2,3\} \times \{1,11,21\}$*, the Krein-level analogues of* (15) *and* (16) *hold bit-exact:*

(i) The geometric Lorentzian modular Hamiltonian $K_L^{\alpha,W} = J_{\text{polar}}|_{W_L}$ *has integer spectrum on the Krein hemispheric wedge.*

(ii) The Krein-Tomita modular operator Δ_L *admits a polar decomposition* $S = J_{L,\text{TT}} \cdot \Delta_L^{1/2}$*; the Tomita modular Hamiltonian* K_L^{TT} *has integer spectrum.*

(iii) $\sigma_{2\pi}^{L,\alpha}(O) = O$ *and* $\sigma_{2\pi}^{L,\text{TT}}(a) = a$ *bit-exact, with cross-witness collapse residual zero.*

(iv) At $N_t = 1$*, the Lorentzian construction reduces to the Riemannian one of Theorems 6.1–6.2 bit-identically.*

Signature-independence of the load-bearing scope. The deepest structural finding of Paper 43 is that the "$H_{\text{local}} \neq D_W^L$" statement is *signature-independent* at the Riemannian reduction. At $N_t = 1$ the residual norm $\|H_{\text{local}} - D_W^L\|_F$ on the Lorentzian Krein wedge equals the Riemannian-side value at every $n_{\max}$ (Paper 43 §7.1):

$$\|H_{\text{local}}|_{(3,1),N_t=1} - D_L^W|_{N_t=1}\|_F \;=\; \|H_{\text{local}}^{\text{Riem}} - D_W\|_F \qquad (\text{at every } n_{\max}).$$

At $N_t > 1$ the Lorentzian residual is refined upward by the temporal-derivative content of D_L: the spectral-action object D_L has more content than D_{GV}, so the gap between H_{local} (modular-Hamiltonian generator) and D_W^L (spectral-action object) widens with temporal cutoff. This refines the Paper 42 §10 open question O3 from "open about signature-dependence" to a deeper structural feature.

6.3 Pythagorean HS-orthogonality and the $1/\pi^2$ signature

Paper 43 §10.2 records a Pythagorean refinement of the modular vs spectral-action gap. Let $\langle \cdot, \cdot \rangle_{\text{HS}}$ denote the Hilbert–Schmidt inner product on the wedge operator algebra.

Corollary 6.4 (Paper 43 §10.2, HS-orthogonality on the Lorentzian wedge)**.** *For every* $\kappa_g > 0$ *and every panel cell* $n_{\max} \in \{1, \ldots, 6\}$*,* $N_t \in \{1, 11\}$*, and witness* $w \in \{\text{BW}, \text{HH}_1, \text{HH}_2, \text{Sew}, \text{Unruh}_1, \text{Unruh}_2\}$*:*

$$\langle H_{\text{local}}, D_W^L \rangle_{\text{HS}} \;=\; 0 \qquad (\textit{bit-exact}), \tag{18}$$

and the residual norm has the closed form

$$\|H_{\text{local}} - D_W^L\|_F^2 \;=\; \frac{\kappa_g^2}{4\pi^2}\, S(n) + D(n), \quad S(n) = \frac{n(n+1)(n+2)(2n^2+4n-1)}{15}, \quad D(n) = \frac{n(n+1)(n+2)(2n}{20} \tag{19}$$

The closed form (19) is PSLQ-verified at 100 digits with coefficient ceiling 10^6 at 18 panel cells. The $1/\pi^2$ prefactor on $\|H_{\text{local}}\|^2$ is the same M_1 Hopf-base measure signature (9) that appears in the Paper 38 GH convergence rate constant $4/\pi$. This crosswiring between sections 4 and 6 is taken up systematically in §8.

Subspace decomposition. The mechanism is the orthogonal decomposition of the operator algebra $B(\mathcal{K}_W)$ into a diagonal subspace (where $H_{\text{local}} = K_\alpha^W/\beta$ lives, since K_α^W has eigenvalues two m_j on the full-Dirac basis) and an off-diagonal subspace with $\Delta n = \pm 1$ (where D_W^L lives, since the Camporesi–Higuchi Dirac intertwines $\pm m_j$ chirality partners at adjacent shells). Formal operator-theoretic proof of this decomposition is named as Paper 43 open question O4; the PSLQ-verified Pythagorean closed form (19) is empirically sharp at 100 digits and 18 panel cells.

7 Lorentzian propinquity

7.1 Operator-system substrate (Paper 44)

The first step in any Lorentzian propinquity proof is an operator- system substrate. Paper 44 [37] provides it on the chirality-doubled Camporesi–Higuchi spinor space $\mathcal{K}_{n_{\max},N_t} = \mathcal{H}_{\mathrm{GV}}^{n_{\max}} \otimes \mathbb{C}^{N_t}$ at BBB signature $(m,n) = (4,6)$.

Theorem 7.1 (Paper 44, propagation number and envelope dependence)**.** *The truncated Lorentzian operator system $\mathcal{O}^L_{n_{\max},N_t}$ satisfies*

$$\mathrm{prop}(\mathcal{O}^L_{n_{\max},N_t}) \;=\; 2 \quad \textit{under the Weyl-doubled achievable envelope}, \quad \mathrm{prop}(\mathcal{O}^L_{n_{\max},N_t}) \;=\; \infty \quad \textit{under the full (di} \tag{20}$$

generalising the Riemannian prop $= 2$ *result for* $\mathcal{O}_{n_{\max}}$ *(Paper 32 §3.4) to the Lorentzian Krein setting. The witness pair* $(M^{2,1,0,0}, M^{2,1,0,0*})$ *at* $n_{\max} = 2$ *exhibits* $\sim 38\%$ *Frobenius residual against* $\mathcal{O}^L_{2,3}$*. Riemannian-limit recovery at* $N_t = 1$ *is bit-exact (Frobenius residual* $= 0.0$ *in float64) across* $n_{\max} \in \{1,2,3\}$*.*

K^+-positivity is trivial at the operator-multiplier level. A key structural observation of Paper 44 is that all chirality- doubled scalar multipliers $M \oplus M$ commute with $J_L =$ chirality-swap, and therefore preserve $\mathcal{K}^+$ automatically: $\mathcal{O}^{L,+} = \mathcal{O}^L$ exactly. Consequently the non-trivial Krein-positivity program shifts from operator level to *state* level (Wasserstein–Kantorovich on K^+ states), which is the natural setting for the Paper 45 propinquity proof below.

7.2 K^+-restricted weak-form Lorentzian propinquity: the degeneracy theorem (Paper 45)

Paper 45 [38] closes off the Lorentzian-propinquity route via a degeneracy theorem (the natural K$^+$-weak-form seminorm vanishes identically) on the substrate of Paper 44.

Definition 7.2 (Paper 45 §1.3, K$^+$-weak-form Lorentzian propinquity)**.** For two truncated Lorentzian spectral triples $\mathcal{T}_i^L$ $(i = 1,2)$, define

$$\Lambda^L(\mathcal{T}_1^L, \mathcal{T}_2^L) \;:=\; \Lambda(\mathcal{T}_1^L|_{\mathcal{K}^+}, \mathcal{T}_2^L|_{\mathcal{K}^+}), \tag{21}$$

where Λ is the standard Latrémolière propinquity [14, 15] and the restriction to $\mathcal{K}^+ = \{|\psi\rangle : J_L|\psi\rangle = +|\psi\rangle\}$ is the standard metric spectral triple obtained from the Krein-positive subspace, on which the indefinite Krein product reduces to a positive-definite Hilbert inner product.

Theorem 7.3 (Paper 45, annihilation theorem)**.** *The device of Definition 7.2 is degenerate. For* D_L *as in* (17)*, Krein-self-adjointness of the anti-Hermitian spatial block forces* $\{J_L, D_{\mathrm{GV}} \otimes I\} = 0$*, hence* $P^+(D_{\mathrm{GV}} \otimes I)P^+ = 0$*: the compression annihilates the spatial Dirac. The surviving temporal block is momentum-diagonal and commutes with every momentum-diagonal temporal multiplier, so the restricted Lipschitz seminorm vanishes identically on the operator system and* Λ^L *of* (21) *is not a quantum Gromov–Hausdorff-type distance. (Bit-exact: restricted-seminorm kernel* $42/42$ *and* $275/275$ *multipliers at* $(n_{\max}, N_t) = (2,3), (3,5)$*; frozen falsifier in the repository test suite.)*

An earlier convergence claim through this device (formerly Theorem 5.1 of Paper 45) is withdrawn. The rate expression

$$C_3^{\mathrm{joint}} \cdot \gamma^{\mathrm{joint}}_{n_{\max},N_t,T} \;=\; O\big(\log n_{\max}/n_{\max} \;+\; T/N_t\big) \tag{22}$$

survives as a well-defined closed form whose meaning is an unconditional *spatial* statement (Paper 38), the temporal factor being a metrically invisible spectator.

7.3 Three structural simplifications

The original transfer of the Paper 38 five-lemma chain to the joint $\mathrm{SU}(2)\times\mathrm{U}(1)$ setting recorded three structural "simplifications." Under the degeneracy analysis each is reinterpreted: they are *symptoms* of the temporal factor's metric invisibility, not conveniences:

PURE_TENSOR Berezin map. The joint Berezin map $B^{\mathrm{joint}} = B^{\mathrm{SU}(2)} \otimes B^{\mathrm{U}(1)}$ factorises as a pure tensor. Each of the four required properties (positivity, contractivity, approximate identity, L3 compatibility) inherits factor-by-factor; no cross-factor verification is needed.

Vanishing time-chirality cross-term. The joint Lichnerowicz commutator simplifies via the structural identity

$$[D_L,\, a_s \otimes a_t] \;=\; i\,[D_{\mathrm{GV}}, a_s] \otimes a_t \tag{23}$$

because $\{\gamma^0, D_{\mathrm{GV}}\} = 0$ on the chirality-doubled basis. (γ^0 anticommutes with the spatial Camporesi–Higuchi Dirac because D_{GV} is chirality-diagonal in the Peskin–Schroeder chiral basis.) The temporal direction contributes nothing to the joint commutator, and $C_3^{\mathrm{joint}} = C_3^{\mathrm{SU}(2)}$ verbatim. Read correctly, (23) is the *diagnosis*: the momentum-diagonal temporal algebra is invisible to the Dirac commutator.

Corrected cb-norm convention. The smoothing map of the joint central spectral Fejér kernel is unital completely positive, with cb-norm exactly 1; the quantity $2/(n_{\max}+1)$ formerly quoted as the "joint cb-norm" is the maximum of the SU(2) Plancherel *mass distribution*, a different invariant (the Bożejko–Fendler appeal is withdrawn; the correct statement is elementary). The N_t-dependence of the rate (22) lives entirely in the joint mass-concentration moment (the T/N_t Fejér-on-the-circle rate).

Riemannian-limit recovery at $N_t = 1$ bit-exact. At $N_t = 1$, the joint Berezin reduces bit-identically to $B^{\mathrm{SU}(2)} \otimes B^{\mathrm{U}(1)}_{1,T} = B^{\mathrm{SU}(2)}$, since the $N_t = 1$ temporal Berezin map is the identity on the single Fourier mode. Hence $\Lambda^L(\mathcal{T}^L_{n_{\max},1,T}, \mathcal{T}^L_{\mathcal{M}})$ agrees with the Riemannian Paper 38 bound bit-exactly. Under the degeneracy analysis this recovery is an automatic formula identity (the panel quantity only ever contained the spatial rate formula), not an operator-level falsifier; the falsifier with teeth is the restricted-seminorm check of Theorem 7.3.

7.4 Honest scope

Theorem 7.3 closes the natural device; the Lorentzian convergence question itself remains open *at the continuum / non-compact level.* The *finite-cutoff* metric question is now understood structurally (Paper 45 § open Q1, 2026-06): the truncated Bisognano–Wichmann boost is *compact* (the modular generator has integer spectrum, $e^{2\pi i K} = I$ at every finite cutoff $\Rightarrow$ the modular flow is the 2π-periodic KMS circle, not a non-compact boost). A compact circle carries only a Euclidean (forward-triangle) metric — which is why the forward chain inequality is the only universal direction and every Lipschitz / boost seminorm is signature-blind — so the finite-cutoff quantum metric is Euclidean and the Lorentzian signature is the Wick rotation $t \to it$ of that circle, a convention rather than a finite-cutoff metric. The genuine reverse-triangle Lorentzian signature requires the non-compact continuum ($n_{\max} \to \infty$, type-III) limit, which is the sense in which the question remains open. This is the metric-level dual of the WH7 reading (compactness $\Rightarrow$ discreteness $\Rightarrow \pi/\beta$). Several aspects of scope require explicit comment.

(Q1) **Strong-form Lorentzian propinquity (Paper 45 §1.4 G1).** The K^+-restriction reduces the indefinite Krein product to a Hilbert inner product on $\mathcal{K}^+$ — but annihilates the spatial Dirac (Theorem 7.3), so no quantum-metric machinery applies on the compression.

Strong-form Lorentzian propinquity (Latrémolière-style metric on Krein spectral triples in their own right, without K^+-restriction) requires an indefinite-inner-product analog of the operator-norm Lipschitz seminorm with appropriate behaviour under Krein-self-adjoint D_L. This remains a substantial open NCG-math problem.

(G2) **De-compactification $T \to \infty$.** The temporal factor is compactified to $\mathrm{U}(1)_T$ with finite radius T. De-compactification to non-compact $\mathbb{R}_t$ is a separate program (internal Sprint L3c) requiring renormalisation of the cb-norm bound and reach moment.

(G3) **Cross-manifold $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$.** Blocked at the NCG-framework level by the Riemannian-vs-Hardy-sector asymmetry recorded in Paper 24 §V of the GeoVac corpus (a four-layer Coulomb/HO asymmetry); this is the Coulomb-side claim only.

(G4) **Inner-factor calibration data.** Higgs / Yukawa selection question of the Connes–Marcolli SM construction; orthogonal to the present convergence theorem, addressed by Paper 32 §8.C verdict POSITIVE-THIN (construction admits Higgs structurally, but no autonomous Yukawa selection from the present data).

8 The master Mellin engine and the cross-arc connection

The most striking structural observation that emerges from reading the arc is that the same transcendental signatures appear in *different* settings. The constant $4/\pi$ of Paper 38 and the constant $1/\pi^2$ of Paper 43 §10.2 are not numerical coincidences: they are two appearances of the same $k = 0$ volume mechanism (historically named "Hopf-base measure"), the M_1 sub-case of the master Mellin engine catalogued in Paper 32 §8.

8.1 The π-source case-exhaustion theorem (Paper 32 §8)

Theorem 8.1 (Paper 32 §8, π-source case-exhaustion). *Let $\mathcal{C}$ be any finite composition of GeoVac projections applied to a Layer-1 datum on $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ or its Bargmann–Segal sibling triple. The following are equivalent:*

(i) $\mathcal{C}$(datum) contains a transcendental of the form π^k for $k \in \mathbb{Q}$;

(ii) $\mathcal{C}$ contains at least one projection whose evaluation includes a continuous integration over a parameter promoted from the discrete graph spectrum;

(iii) $\mathcal{C}$ engages at least one of three spectral-triple mechanisms,

M_1 *(*Hopf-base measure*): $\pi = \mathrm{Vol}(S^2)/4$ via the Hopf bundle $S^1 \to S^3 \to S^2$, equivalently $4/\pi = \mathrm{Vol}(S^2)/\pi^2$;*

M_2 *(*Seeley–DeWitt*): $\sqrt{\pi}$ via heat-kernel coefficients on the Camporesi–Higuchi Dirac spectrum on S^3 (Paper 28 Theorem T9: $\zeta_{D^2}(s)$ is polynomial in π^2 with rational coefficients at every integer s);*

M_3 *(*vertex-topology Dirichlet L*): Catalan G and $\beta(s)$ via quarter-integer Hurwitz shifts in two-loop sunset vertices.*

The TS-E3 sharpening of Paper 32 further shows that M_1, M_2, M_3 are sub-cases of a single master Mellin-engine mechanism

$$\pi\text{-source} \;=\; \mathcal{M}\big[\mathrm{Tr}(D^k e^{-tD^2})\big] \quad \text{with } k \in \{0, 1, 2\}, \tag{24}$$

where $k = 0$ selects M_1, $k = 1$ selects M_3, and $k = 2$ selects M_2.

8.2 M_1 in the Riemannian GH rate

The rate constant $4/\pi$ in (6) carries the M_1 ($k = 0$ volume) signature. This is not a derived observation; it is forced by the Plancherel-weight × Vandermonde-Jacobian cancellation theorem (Paper 40 §3.2) which leaves only the rank-1 (SU(2)) constant after the $\prod_{\alpha>0}$ products cancel. Equivalently, the single-positive-root Euler–Catalan sum

$$\sum_{d \text{ odd}} \frac{1}{d^2} = \frac{\pi^2}{8}$$

produces the π^2 in the denominator, and the Hopf bundle $S^3 \to S^2$ produces $\mathrm{Vol}(S^2) = 4\pi$ in the numerator.

8.3 M_1 in the Lorentzian HS-orthogonality

The $1/\pi^2$ prefactor on $\|H_{\text{local}}\|^2$ in the Pythagorean closed form (19) is the same M_1 signature *at the operator-system level*: the Krein wedge BW choice $H_{\text{local}} = K_\alpha^W/(2\pi)$ introduces a factor of $1/(2\pi)$ from the modular-circle circumference, which squared gives $1/(4\pi^2)$ in $\|H_{\text{local}}\|^2$. The volume factor $4/\pi$ of Paper 38 and the $1/\pi^2$ of Paper 43 are two manifestations of the same $\mathrm{Vol}(S^2) = 4\pi$ identity in the master Mellin engine.

Why this matters. The case-exhaustion theorem (Theorem 8.1) is a *spectral-triple-level* statement about which transcendentals can appear in evaluations of GeoVac projection chains. The constant $4/\pi$ of Paper 38 is a *metric-geometry-level* statement about the state-space GH rate. The fact that these two abstractly-different levels agree on the same signature is non-trivial: it identifies the metric geometry of the GeoVac spectral triple as carrying the same M_1 signature that the abstract Mellin engine catalogues. Abstract taxonomy meets concrete state-space GH rates. Paper 43 §10.2 takes this further: the Lorentzian Pythagorean decomposition at the operator-system level carries the same M_1 signature again, this time on the operator algebra of the Krein wedge.

9 Closures since drafting (May 23, 2026)

The synthesis above (§1–§8) was drafted around the arc Papers 29, 32, 38, 39, 40, 42, 43, 44, 45. On 2026-05-23, four substantive closures landed in a single session, adding Papers 46 and 47 to the arc and promoting two named open follow-ons within Papers 39 and 43. This section records the four closures in summary form.

9.1 Strong-form Lorentzian propinquity (Paper 46)

Status: the strong-form convergence claim of this subsection is withdrawn. On the natural chirality-doubled substrate the operator-norm Lipschitz seminorm L_{op} has kernel far exceeding the scalars (every temporal multiplier $\mathbf{1} \otimes M^{\text{temp}}$ is L_{op}-invisible), so neither side of (25) is a quantum-metric distance between quantum compact metric spaces; the bit-exact panel match below re-evaluates the same spatial rate formula and is not an independent convergence. The enlarged-substrate algebra of Appendix B survives and is the named starting point for the repair.

Paper 45 §1.4 G1 named the strong-form Lorentzian propinquity (a Latrémolière-style metric on Krein spectral triples without K^+-restriction) as the principal open NCG-math problem of the Lorentzian arc. Paper 46 [39] *claimed to close* this question on the natural chirality-doubled scalar-multiplier substrate of Paper 44 [37] via the operator-norm Lipschitz seminorm $L_{\text{op}}(a) := \|[D_L, a]\|_{\text{op}}$:

$$\Lambda^{\text{strong}}(\mathcal{T}^L_{n_{\max},N_t,T},\ \mathcal{T}^L_{\mathcal{M}}) \ \le\ C_3^{\text{op}}(n_{\max}) \cdot \gamma^{\text{joint}}_{n_{\max},N_t,T} \ \to\ 0, \qquad (25)$$

with operator-norm constant $C_3^{\rm op}$ — for which the earlier closed form $\sqrt{1-1/n_{\max}}$ is *withdrawn* as operator-norm-false (the correct comparison is the Paper 38 Lemma L3 value $C_3 = 1$; see Paper 46 App. A.3) — and rate $\gamma^{\rm joint} = O(\log n_{\max}/n_{\max} + T/N_t)$ from Paper 45. The numerical panel matches Paper 45's K^+-weak-form panel *bit-exactly* at every tested cell — the "free-upgrade" reading.

Paper 46's Appendix B (added 2026-05-22) extends the closure to the *enlarged* operator system with chirality-flipping generators satisfying $\{J_L, M^{\rm flip}\} = 0$. The substantive new content is the *gradient-norm absorption mechanism*: the strict-strong-form separation between enlarged and K^+-weak-form substrates manifests at the Lipschitz seminorm level ($L_{\rm op}(a^{\rm flip}) > 0$ on enlarged generators while $L^+_{\rm P45}(a^{\rm flip}) = 0$ identically), but the separation is paid *entirely in a J-graded gradient-norm extension* $G^{\rm enlarged} = G^{\rm natural} + 2\|D_t\|_{\rm op}\|f^{\rm flip}\|_{\rm op}$, leaving the Latrémolière propinquity bound bit-exactly preserved: $\Lambda^{\rm enlarged}(n_{\max}, N_t) = \Lambda^{\rm P46}(n_{\max}, N_t)$ at every panel cell.

9.2 De-compactification limit, norm-resolvent / spectral closure (Paper 47)

Status: the outer *norm-resolvent arrow of* (26) *and the three-carrier identification stand; the* inner *propinquity arrow (Paper 45/46 main theorem applied along the scaling) is descoped by Paper 45's degeneracy theorem, so the composite is established at the norm-resolvent / spectral level only.*

Paper 45 §1.4 G2 named the de-compactification limit $T \to \infty$ on the temporal carrier S^1_T as a separate open program. Paper 47 [40] closes it at the *norm-resolvent / spectral* level via a *two-rate hybrid composite*:

$$\mathcal{T}^L_{n_{\max},N_t,T(N_t)} \xrightarrow{\Lambda^L\text{-rate } O(\log n_{\max}/n_{\max} + T(N_t)/N_t)} \mathcal{T}^L_{S^3\times S^1_{T(N_t)}} \xrightarrow{\text{nr-rate } O(e^{-|\Im z| T(N_t)/2})} \mathcal{T}^L_{S^3\times\mathbb{R}_t} \tag{26}$$

along any *admissible scaling* $T(N_t) \to \infty$ with $T(N_t)/N_t \to 0$. The inner arrow is propinquity convergence at coupled cells (Paper 45/46 main theorem applied along the scaling); the outer arrow is norm-resolvent convergence of the Lorentzian Dirac on the compactified spacetime to the Lorentzian Dirac on the genuine spacetime, with explicit exponential tail bound $O(e^{-|\Im z|T/2})$ on compact-support subspaces via Reed–Simon [41] § XIII.16 / Bloch-decomposition machinery and the Krein-isometric embedding $\widetilde{J}_T : \ L^2(S^1_T) \hookrightarrow L^2(\mathbb{R})$.

A strategic clarification appears in Paper 47 §1.1: the natural *inductive-limit propinquity* framework of Latrémolière [15] § 5 for AF C*-algebras is structurally wrong for the present setting because AF C*-algebras have totally-disconnected Gelfand spectrum while $\mathbb{R}$ is connected, so $C_0(\mathbb{R})$ cannot be an AF inductive limit. Norm-resolvent convergence is the right framework at the spectral level.

Paper 47 §6 also establishes a *three-carrier identification*: the periodic compact $S^3 \times S^1_T$ (Paper 45 substrate), the bounded Dirichlet $S^3 \times [-T/2, T/2]$ (Paper 43 / Sprint L2-B/E substrate), and the non-compact $S^3 \times \mathbb{R}_t$ (Paper 47 limit) all converge in norm-resolvent sense to the same physical Lorentzian Dirac as $T \to \infty$. This resolves the apparent tension between Paper 43's bounded-interval architecture (chosen to preserve causal structure / no CTCs via Geroch's theorem) and Paper 45's periodic-compact architecture (chosen to enable Latrémolière propinquity machinery on a compact carrier). Both are valid compact approximations to the same physical Lorentzian Dirac.

A numerical-verification panel (Sprint L3c-α.2, 2026-05-23) at 24 cells $(n_{\max}, N_t) \in \{2,3\} \times \{3,5,7,11\}$ along three scalings ($T_0 = 2\pi$ canonical, $T_1 = N_t/\log N_t$ sub-linear, $T_2 = \sqrt{N_t}$ square-root) confirms Λ^L bit-exact T-and-N_t-independence at fixed $n_{\max}$: $\Lambda(n_{\max} = 2) = 2.074551$ and $\Lambda(n_{\max} = 3) = 1.610060$ to 6+ significant figures across all three scalings, matching Paper 45 Table 1 verbatim. Structural reason: $\gamma^{U(1)}(N_t, T)$ is subordinate to $\gamma^{\rm SU(2)}(n_{\max})$ at every admissible-scaling cell, so the joint rate $\gamma^{\rm joint} = \max(\gamma^{\rm SU(2)}, \gamma^{U(1)}) = \gamma^{\rm SU(2)}$ exactly.

The empirical confirmation upgrades the L3c-α theorem from "analytical corollary of Paper 45" to "analytically AND empirically established."

9.3 Pythagorean orthogonality formal proof (Paper 43 §10.2 promotion)

Paper 43 §11 O4 named a formal proof of the diagonal/off-diagonal subspace-decomposition mechanism behind the L2-F.1 (2026-05-17) empirical observation $\langle H_{\text{local}}, D_L^W \rangle_{\text{HS}} = 0$ as a follow-on. The formal proof landed on 2026-05-23 via a $\mathbb{Z}_2$ wedge-chirality parity argument and is now recorded in Paper 43 §10.2 as a formal theorem.

The argument has three structural inputs from Papers 42/43/44 [35–37]: (I1) the wedge chirality grading Π_W (BBB Krein-chirality γ^5 restricted to the wedge) is a unitary self-adjoint involution; (I2) $[\Pi_W, H_{\text{local}}] = 0$ (even parity of the geometric BW-α generator $K_\alpha^W = J_{\text{polar}}^W$ under wedge chirality); (I3) the wedge-restricted Lorentzian Dirac D_L^W has *mixed* parity — its temporal factor is Π_W-*odd* ($\{\gamma^0, \gamma^5\} = 0$) while its spatial factor D_{GV}^W is Π_W-*even* but chirality *sign-paired* (D_{GV} is chirality-diagonal — the same fact that makes the BBB universal axiom $\{\chi, D\} = 0$ *fail* on the truthful D_{GV} — so $D_{\text{GV}}^W = \Pi_W |D_{\text{GV}}^W|$ with $|D_{\text{GV}}^W|$ even and χ-independent).

The orthogonality follows from two cyclicity-of-trace lemmas: (a) for any unitary involution Π, $[\Pi, A] = 0$ and $\{\Pi, B\} = 0$ imply $\text{Tr}(A^\dagger B) = 0$ via $\text{Tr}(A^\dagger B) = \text{Tr}(A^\dagger \cdot (-B)) = -\text{Tr}(A^\dagger B)$ (this kills the temporal factor); (b) for A even and χ-independent and $B = \Pi|B|$ even and sign-paired, $\text{Tr}(A^\dagger B)$ cancels pairwise across the $\chi = \pm 1$ partners (this kills the spatial factor). The bit-exact L2-F.1 numerical panel at 18 cells is empirical corroboration of the now-algebraic identity at machine precision. (An earlier one-line "D_L^W is Π_W-odd" statement was corrected to this mixed-parity form, 2026-06-23, after the chirality-pairing test showed D_{GV}^W is chirality-diagonal hence Π_W-even.)

9.4 Master theorem of the math.OA arc

Paper 39 §6 (iii) and (v) named two open follow-ons: (iii) the k-fold extension of the Paper 39 two-factor tensor-product propinquity to $\mathcal{T}_{S^3}^{\otimes k}$, and (v) the higher-rank tensor product $\mathcal{T}_{G_1} \otimes \mathcal{T}_{G_2}$ at arbitrary compact connected Lie group factors. Both closed on 2026-05-23.

(iii) k-fold extension — $\sqrt{k}$ constant (sketch). The k-fold extension to SU(2) factors uses the graded Leibniz rule and the operator-norm *triangle* bound, giving $C_3^{(k)} \leq \sqrt{k}\,\sqrt{(N_*-1)/(N_*+1)} \nearrow \sqrt{k}$ where $N_* := \max_j n_j$. *Correction (2026-06-18):* an earlier draft claimed a k-**independent** constant $\sqrt{(N_*-1)/(N_*+1)} \to 1^-$ via a "Pythagorean Leibniz" sum-of-squares identity $\|\sum_j X_j\|_{\text{op}}^2 = \sum_j \|X_j\|_{\text{op}}^2$. That identity is **false** (pairwise anticommutation of the $D^{(k)}$ summands does not make the commutator terms operator-norm orthogonal — verified maximally violated at $k = 2$ on the actual harmonics). The genuine $\sqrt{k}$ factor is real, not eliminated; for each fixed k convergence still holds (finite constant), but the constant grows with the number of factors. Only the $k = 2$ case is proved (Paper 39); $k > 2$ is a sketch.

(v) Master theorem — universal compact-Lie-group factors (sketch). Combining the k-fold extension with the universal $4/\pi$ rate of Paper 40 [34] gives the expected multi-factor form

$$\Lambda^{(k)}\big(\mathcal{T}^{(k)}_{n_1,\ldots,n_k},\ \mathcal{T}^{(k)}_{G_1\times\cdots\times G_k}\big) \ \leq\ C_3^{(k)} \cdot \max_j \gamma_{n_j}^{G_j}, \qquad C_3^{(k)} \leq \sqrt{k}\,\sup_j C_3^{G_j}(n_j) \nearrow \sqrt{k}, \tag{27}$$

for arbitrary $k \geq 1$ and compact connected semisimple Lie groups $G_1, \ldots, G_k$ with bi-invariant metrics (Paper 40's class; a central torus factor is excluded — Paper 40 asserts no value for $c(G_{\text{ss}} \times T^k)$). *Correction (2026-06-18):* this was boxed as a closed "master theorem" with a k-**independent** constant $\sup_j C_3^{G_j}(n_j)$; with the corrected triangle bound the constant carries a genuine $\sqrt{k}$ and is not k-independent. The joint rate then inherits the $4/\pi$ asymptotic; but this multi-factor / general-G form is a **sketch** (only the $k = 2$, SU(2) and $k = 1$ cases are proved).

The master theorem subsumes Papers [32] ($k = 1$, $G = \mathrm{SU}(2)$), [34] ($k = 1$, general G), and the two-factor case treated above (§4.8, $k = 2$, $G_1 = G_2 = \mathrm{SU}(2)$) as special cases. Paper 45's product-carrier proposition at $S^3 \times S^1_T$ (`prop:product_action_seminorm`; signature-agnostic and not a Lorentzian claim — its earlier compression-device convergence theorem is withdrawn) is a $k = 2$, $G_1 = \mathrm{SU}(2)$, $G_2 = \mathrm{U}(1)$ product with per-factor cutoffs whose joint rate is the maximum of the factor rates, i.e. the SU(2) rate (§4.8); it lies outside Paper 40's semisimple class and is not an instance of the master theorem's universal constant.

10 Further closures: Krein–Mondino–Sämann bridge, OSLPLS, and bit-exact CFT-on-sphere

The arc admits three further closures landed subsequent to the May 22, 2026 drafting of this synthesis. These add three further papers (48, 49, 50) and extend the math.OA standalone count from eleven to fourteen.

10.1 Krein-pointed proper QMS and the Mondino–Sämann bridge (Paper 48)

Status: the metric-level content of this subsection is descoped by Paper 45's degeneracy theorem; the bridge design survives as a conditional proposal.

Paper 48 [42] constructs Krein-pointed proper quantum metric spaces $\mathbf{KreinMetaMet}_{\mathrm{pp}}$ as the natural Krein-lift of Latrémolière's recent pointed proper quantum metric space framework [51], and builds a *bridge functor* $W : \mathbf{KreinMetaMet}_{\mathrm{pp}} \to \mathbf{LorPLG}_{\mathrm{cov}}$ to the Mondino–Sämann synthetic Lorentzian pre-length space category [18, 19]. The bridge is identified via the Connes–Rovelli thermal-time hypothesis [50]: the modular distance ^{K} on the Krein side measures thermal time (additive, forward triangle), while the proper-time distance ℓ^L on the Mondino–Sämann side measures geometric proper time (super-additive, reverse triangle); Connes–Rovelli identifies these as two readings of the same wedge KMS state, related by Wick rotation. The bridge is functorial, not isometric.

Conditional on the bridge construction — whose K^+-weak-form input is descoped by Paper 45's degeneracy theorem — Bridge Theorem 6.4 with all four properties (B1)+(B2)+(B3)+(B4) would yield seven newly accessible theorems on the GeoVac hemispheric wedge, including: (T3) bit-exact panel inheritance — the $\Lambda(2, 3) = 2.0746$, $\Lambda(3, 5) = 1.6101$, $\Lambda(4, 7) = 1.3223$ panel of Paper 45 would transport verbatim; (T5) synthetic Pythagorean orthogonality with the $1/\pi^2$ master Mellin engine M_1 signature as the inverse-squared bridge cover slab-period; and (T6) a proposed G2 metric-level *route* for the GeoVac wedge specifically, via the bridge $\to$ MS pLGH workaround. This last item is the most substantive design content of Paper 48: it offers a conditional synthetic-side workaround for the Paper 45 §1.4 G2 metric-level problem (contingent on the descoped bridge, hence a proposed route rather than a closure), sidestepping the absence of a published non-compact Latrémolière propinquity by working through the Mondino–Sämann category instead.

Paper 48 is the tenth math.OA standalone in the arc, sibling to Papers 38, 42, 43, 44, 45, 46, 47, 29, 32, 40.

10.2 OSLPLS and the strong-form bridge: twin paradox as quantum information (Paper 49)

Status: the Λ-inheritance and K^+-framed claims of this subsection are descoped by Paper 45's degeneracy theorem; the cocycle-deficit algebra and the OSLPLS category design survive.

Paper 49 [43] closes Paper 48 §8.1's open question Q1′ by constructing an *OSLPLS* (Operator-System Lorentzian Pre-Length Space) category and a strong-form bridge functor $W^{\mathrm{flip}} : \mathbf{KreinMetaMet}^{\mathrm{flip}}_{\mathrm{pp}} \to \mathbf{OSLPLS}_{\mathrm{cov}}$ on the Paper 46 Appendix B enlarged substrate (chirality-flipping generators $M^{\mathrm{flip}} = \mathrm{diag}(W, -W)$ where $\{J, M\} = 0$). The closure rests on a genuinely new operator-algebraic construction: the Connes–Rovelli thermal-time stack across *distinct* KMS states, extending Connes–Rovelli [50] from the single-KMS-state setting to multi-state stack consistency. Properties (B1′)+(B2′)+(B3′) hold theorem-grade; (B4′), the metric-level Λ-inheritance, is *descoped* (the Krein-side seminorm is degenerate, Paper 45/46).

The headline structural finding is the *twin-paradox-as-quantum-information* reading: strict super-additivity of the OSLPLS reverse triangle on off-orbit triples is the operator-algebraic dual of the special-relativistic twin paradox, with the deficit quantified by Datta's max-divergence chain inequality [52]. The detoured worldline pays *two* max-divergence cocycle costs $\Delta_{\max} S^{\sigma_1 \to \sigma_2} + \Delta_{\max} S^{\sigma_2 \to \sigma_3}$; the direct worldline pays *one* cost $\Delta_{\max} S^{\sigma_1 \to \sigma_3}$; the strict inequality follows from Datta 2009's max-divergence chain inequality (the Umegaki relative entropy itself does not satisfy a chain inequality in general — see Paper 49 §5.4 and Q3′ for the honest scope). Numerical verification at $(n_{\max}, N_t) \in \{(2,3), (3,5), (4,7)\}$ confirms: Λ bit-exact at all three panel cells (rate-formula consistency, not a metric convergence), *illustrative* Umegaki relative-entropy deficits 66.998 / 68.720 / 81.256 nats positive and monotone-increasing in $n_{\max}$ (the load-bearing $D_{\max}$ chain is verified 96/96), Riemannian-limit recovery at $N_t = 1$ bit-exact.

The Bousso–Chandrasekaran–Rath–Shahbazi-Moghaddam 2020 "kink transform" / Connes cocycle flow gravity dual [49] appears here as a natural bulk reading of the OSLPLS construction (Paper 49 §11): the strict super-additivity of the OSLPLS reverse triangle on off-orbit triples translates, under the kink-transform dictionary, to a positive relative entropy on the gravitational dual — recovering the BCFM positivity statement from the operator-algebraic side.

Paper 49 is the eleventh math.OA standalone in the arc. Closed-form expression for the cocycle entropy production deficit as a function of OSLPLS state-space coordinates remains a named open question (Paper 49 §10); the genuine non-commutative Mondino–Sämann pre-length-space extension is the separate named open question Q2′, partially addressed by the OSLPLS containment (Paper 49 §10).

10.3 Bit-exact CFT-on-sphere partition functions and the master Mellin engine M_2/M_3 orthogonal decomposition (Paper 50)

Paper 50 [44] demonstrates that the framework's existing Hurwitz-zeta machinery (used throughout Papers 38–47 for spectral sums on the Camporesi–Higuchi tower) reproduces continuum CFT-on-sphere partition functions *bit-exactly*, via a closed-form combination of $\zeta'(-2)$, $\zeta'(-4)$, $\zeta(3)$, $\zeta(5)$, $\log 2$, and π identified by PSLQ at 80–100 digit precision.

Bit-exact match on S^3. For the conformally coupled scalar and Camporesi–Higuchi Dirac on round S^3,

$$F^{S^3}_{\mathrm{scalar}} = \frac{\log 2}{8} - \frac{3\zeta(3)}{16\pi^2} \approx 0.0638 \qquad \text{(PSLQ integer relation } [8, 2, -3]\text{)},$$
$$F^{S^3}_{\mathrm{Dirac}} = \frac{\log 2}{4} + \frac{3\zeta(3)}{8\pi^2} \approx 0.21896 \qquad \text{(symbolic Hurwitz expansion)}.$$

These reproduce the Klebanov–Pufu–Safdi values [46] bit-exactly.

Orthogonal decomposition and log 3 cancellation theorem on S^5. On the (scalar, Dirac) sector, the framework reveals an orthogonal decomposition:

$$F_D + 2F_s = \frac{\log 2}{2} \quad (M_2 \text{ sector}), \qquad F_D - 2F_s = \frac{3\zeta(3)}{4\pi^2} \quad (M_3 \text{ sector}).$$

On S^5, the pattern extends: $F^{S^5}_{\text{scalar}} = -\log(2)/32 - \zeta(3)/(32\pi^2) + 15\zeta(5)/(64\pi^4)$ (PSLQ relation $[32, -2, -2, 15]$), $F^{S^5}_{\text{Dirac}} = -(3\log 2)/128 - 5\zeta(3)/(128\pi^2) - 15\zeta(5)/(256\pi^4)$. The log 3 contributions that arise from zero-mode boundary structure in the minimal-scalar variant cancel algebraically against M_2-sector identities (Paper 50 Theorem 7.2), restricting the substrate-natural ring to $\log 2 \cdot \mathbb{Q} + \zeta(3)/\pi^2 \cdot \mathbb{Q} + \zeta(5)/\pi^4 \cdot \mathbb{Q}$ on S^5.

S^7–S^{11} **ladder closure.** The ladder *generates*: every odd rung $S^3, \ldots, S^{11}$ has a rational-coefficient closed form in $\mathcal{R}_d$, with no transcendental outside $\mathcal{R}_d$ (Paper 50 §8). An earlier reported S^7 non-match was a search-resolution artifact (a 30-digit PSLQ search; the genuine S^7 scalar relation, with integer coefficients of order 10^4, lies in $\mathcal{R}_7$ and appears at ≥ 200 digits with reconstruction error $\approx 2.4 \times 10^{-202}$ — Paper 50 §8, Erratum).

Cataloguing the master Mellin engine in physics. The S^3 and S^5 closures are recorded in Paper 50 §9 as the first two entries of a running catalogue of master-Mellin-engine appearances on the physics side. Paper 50 also flags structural correspondences to other M_2-sector results in the QFT literature, including Henningson–Skenderis holographic Weyl anomaly [47] (M_2 Seeley–DeWitt content), TT-bar / cutoff AdS [48], and the Chamseddine–Connes spectral action [5] (also M_2). These correspondences are flagged as structural, not as new derivations.

Structural reading: scale-freedom on substrate $\leftrightarrow$ scale-invariance in physics. The bit-exact CFT closures on S^3 and S^5 are structurally explained by the framework's substrate-level scale-freedom: the GeoVac packing axiom outputs a graph whose Laplacian eigenvalues are pure rationals $n^2 - 1$, the multiplicities are integers, the Wigner-symbol couplings are exact rationals, and the packing diameter d_0 *cancels out* of every spectral quantity. Physical scales enter only through the Fock projection focal length p_0, which is a stereographic conformal map. This scale-freedom on the substrate is the geometric reflection of the same principle that conformal field theory [46] embodies on the physics side: a CFT has no preferred length scale at the level of its action and observables; the GeoVac graph has no preferred length scale at the level of its eigenvalues and matrix elements. The framework's ability to host continuum CFT-on-sphere observables bit-exactly reflects this structural alignment. This is *not* a claim that GeoVac *is* a CFT; it is a structural reflection that explains *why* the framework reproduces what it does. (The substrate-side account of this principle is developed in the Group 3 foundations synthesis [45].)

Paper 50 is the twelfth math.OA standalone in the arc.

11 Open questions and outlook

The arc closes the convergence side of the operator-algebra story on the Riemannian / spatial side (Paper 38, unconditional) and the modular-Hamiltonian side at the operator-system level on Riemannian (Paper 42) and Lorentzian (Paper 43) backgrounds; the Lorentzian *convergence* question is left open by Paper 45's annihilation theorem. Several questions remain open.

Strong-form Lorentzian propinquity (Q1) — OPEN (Paper 46 descoped, §9.1). Strong-form Lorentzian propinquity — a Latrémolière-style metric on Krein spectral triples in their own right, without K^+-restriction (Paper 45 §1.4 G1) — remains the principal open NCG-math problem of the Lorentzian arc. Paper 46 [39] proposed a closure via the operator-norm Lipschitz seminorm $L_{\mathrm{op}}(a) = \|[D_L, a]\|_{\mathrm{op}}$ on the chirality-doubled scalar-multiplier substrate of Paper 44, but that seminorm has kernel far exceeding the scalars (the temporal multipliers are L_{op}-invisible), so the construction is not a quantum-metric distance and is descoped pending repair; the "free-upgrade" bit-exact panel match is a re-evaluation of the spatial rate formula, not an independent Lorentzian convergence. The enlarged-substrate algebra of Paper 46 Appendix B (chirality-flipping generators with $\{J, M\} = 0$, where the commutators are non-zero) survives and is the named starting point for the repair. A genuine strong-form theorem requires an indefinite-inner-product Lipschitz seminorm compatible with Krein-self-adjoint D_L; a substantial open program.

De-compactification $T \to \infty$ (G2) — CLOSED at norm-resolvent level by Paper 47 (§9.2); G2-metric open as a named frontier. The temporal factor $\mathrm{U}(1)_T$ is finite; the standard physics applications (Hawking, Unruh) require non-compact $\mathbb{R}_t$. The rate $T/N_t \to 0$ in (22) is the joint mass-concentration on $\mathrm{U}(1)_T$ as $N_t \to \infty$ at fixed T; the $T \to \infty$ limit is closed at the *norm-resolvent / spectral* level by Paper 47 [40]'s two-rate hybrid composite (26). The remaining open question is the *metric / propinquity* level closure on a non-compact carrier (G2-metric or equivalently G1-genuine on the non-compact substrate), which requires a non-compact extension of Latrémolière's propinquity not in the published literature as of May 2026. This is named open frontier (Sprint L3e in our internal taxonomy), with the path-of-least-resistance being a pointed-propinquity bridge to the Mondino–Sämann synthetic Lorentzian GH program (§9.2).

Cross-manifold $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$ (G3, Coulomb-side claim only). The natural tensor product of the Coulomb S^3 spectral triple with the Bargmann–Segal Hardy-sector spectral triple on S^5 [29] is blocked at the NCG-framework level by a four-layer asymmetry recorded in Paper 24 §V. The four layers are: (i) spectrum-computing role of L_0 (Coulomb-side only, by Paper 23 Fock rigidity); (ii) calibration π (Coulomb-side only); (iii) non-abelian Wilson gauge with natural matter coupling (both sides have Wilson, only Coulomb-side has natural matter coupling because the Bargmann tower $(N, 0)$ irreps have N-dependent dimensions and no uniform-rep matter sector); and (iv) modular- Hamiltonian structure of the wedge KMS state (Coulomb-side admits the Krein wedge HS-orthogonality of Corollary 6.4 with closed-form M_1 prefactor; the Hardy-sector side does not admit the construction at all). Closing G3 requires new NCG-framework machinery to bridge Riemannian and Hardy-sector spectral triples in a single tensor product; we make no claim about whether this is achievable.

Higgs as inner fluctuation (G4). The GeoVac triple is naturally combined with a finite almost- commutative factor $\mathcal{A}_{\mathrm{GV}} \otimes (\mathbb{C} \oplus \mathbb{H})$ to produce an electroweak-like sector. The Connes axiom audit at finite cutoff (Paper 32 §8.C) gives verdict POSITIVE-THIN: the construction *admits* a Higgs sector structurally (i.e. the inner fluctuation $D \mapsto D + \omega + \varepsilon' J\omega J^{-1}$ produces a non-trivial off-diagonal $\mathbb{C} \leftrightarrow \mathbb{H}$ block iff the finite Yukawa Y_F is non-zero), but *GeoVac does not autonomously select* Y_F from its internal data. In Marcolli–vS terms, GeoVac sits on the Yang–Mills-without-Higgs side of the Perez-Sanchez [21] clarification, with calibration data (the Yukawa) supplied externally. Paper 32 §8.C records this as the standing structural verdict.

State-side complement of the dictionary. Paper 44 highlights that the non-trivial K^+-positivity content of the Lorentzian operator system sits at the *state* level, not the operator level. The natural object is the Wasserstein–Kantorovich metric on K^+-states. Paper 45's K^+-weak-form analysis (the annihilation theorem) is the first probe in this direction and shows the operator-level compression is degenerate; substantially more mathematics on Wasserstein–Kantorovich on Krein-positive *states* is required to complete the state-side dictionary.

Concluding remark. The structural identification of GeoVac as an almost-commutative spectral triple in the Marcolli–van Suijlekom lineage is proved at the state-space Gromov–Hausdorff level by Paper 38 (Riemannian, single factor), extended by Paper 39 (tensor product, state-space GH) and Paper 40 (compact connected semisimple Lie groups). The Lorentzian K^+-weak-form extension is *descoped*: Paper 45 proves a degeneracy theorem (the natural Lorentzian-propinquity seminorm vanishes identically), and what survives on the temporal factor is Paper 45's signature-agnostic product-carrier rate (not a Lorentzian convergence). The metric geometry of the truncated triples carries the same M_1 Hopf-base measure signature ($4/\pi$ in the Riemannian rate, $1/\pi^2$ in the Lorentzian Pythagorean closed form) that the abstract π-source case-exhaustion theorem (Theorem 8.1) identifies at the spectral-triple level. The cross-arc identification of these signatures is the deepest structural observation of the corpus.

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